

WIRE SIZING AND RUNS

Supplying a gate operator with the correct electrical service is crucial to the performance of the operator and the life of its electrical components. If the wire size used is too small, the voltage loss, especially during motor startup, will prevent the motor from attaining its rated horsepower. The percentage of horsepower lost is far greater than the percentage of voltage loss.

A voltage loss can also cause the control components to chatter while the motor is starting, substantially reducing their life due to the resultant arcing. There is no way to restore lost performance resulting from undersized wires, except to replace them. Be sure to choose a sufficient wire size at initial installation to avoid costly rewiring.

The tables on the following page are based on copper wire and allow for a 5% voltage drop. The ampere values shown are the service factor ampere rating of the motor (maximum full load at continuous duty). At minimum, a 20A circuit (protected with a 20A Inverse time Breaker) should be provided.

Always connect electrical power and ground the operator in accordance with the NFPA 780 & NEC, Article 430 and Article 250. Research and adhere to other local codes that may apply.

Low Voltage Control Wiring

The Smart Touch Controller has very sensitive control inputs. The following is a chart of maximum distances for wire size:

Wire Size	Maximum Distance
18 ga	7.0 miles (11 km)
20 ga	3.5 miles (5.6 km)
22 ga	2.7 miles (4.3 km)
24 ga	2.0 miles (3.2 km)
26 ga	1.0 mile (1.6 km)
28 ga	3700 feet (1.1 km)

StrongArm Wiring Charts (Incoming Power)

The maximum distance shown is from the operator to the power source, assuming that source power is from a panel box with adequate capacity to support the addition of this motor load. The values are for one operator, with no other loads applied to the branch circuit. Avoid placing more than one operator to a circuit, but if you must, be certain to reduce the maximum allowed wire distance by half.

NOTE: Distance shown in U.S. Standard "feet." Metric equivalent shown in parentheses.

StrongArm Wire Size Chart – 115V Single Phase				
Horsepower	½	¾	1	2
Amps	10	11.6	14.4	27.2
Wire Gauge	Distance	Distance	Distance	Distance
12	90 (27m)	75 (23m)	60 (18m)	30 (9m)
10	140 (43m)	120 (37m)	100 (30m)	50 (15m)
8	220 (67m)	190 (58m)	155 (47m)	80 (24m)
6	350 (107m)	300 (91m)	245 (75m)	130 (40m)
4	555 (169m)	480 (146m)	385 (117m)	205 (62m)
2	890 (271m)	765 (233m)	620 (189m)	330 (101m)

StrongArm Wire Size Chart – 208V Single Phase					
Horsepower	½	¾	1	2	3
Amps	5.5	6.1	7.6	14.2	16.2
Wire Gauge	Distance	Distance	Distance	Distance	Distance
12	290 (88m)	260 (79m)	205 (62m)	110 (33m)	100 (30m)
10	460 (140m)	415 (126m)	330 (101m)	175 (53m)	155 (47m)
8	725 (221m)	650 (198m)	525 (160m)	280 (85m)	245 (74m)
6	1150 (350m)	1040 (317m)	835 (254m)	445 (135m)	390 (119m)
4	1825 (556m)	1645 (501m)	1320 (402m)	710 (216m)	620 (189m)
2	2920 (890m)	2630 (801m)	2110 (643m)	1130 (344m)	1000 (305m)

StrongArm Wire Size Chart – 230V Single Phase						
Horsepower	½	¾	1	2	3	5
Amps	5.0	5.8	7.2	13.6	14.8	27.0
Wire Gauge	Distance	Distance	Distance	Distance	Distance	Distance
12	350 (107m)	300 (91m)	245 (75m)	130 (40m)	120 (37m)	65 (20m)
10	560 (171m)	480 (146m)	385 (117m)	205 (62m)	190 (58m)	105 (32m)
8	880 (268m)	760 (232m)	610 (186m)	325 (99m)	300 (91m)	165 (50m)
6	1400 (427m)	1120 (341m)	975 (297m)	515 (157m)	475 (145m)	260 (79m)
4	2220 (670m)	1915 (584m)	1550 (472m)	815 (248m)	750 (229m)	410 (125m)
2	3550 (1082m)	3080 (939m)	2465 (751m)	1305 (398m)	1200 (366m)	650 (198m)

NOTE: Distance shown in U.S. Standard "feet." Metric equivalent shown in parentheses.

StrongArm Wire Size Chart – 208V Three Phase						
Horsepower	½	¾	1	2	3	5
Amps	2.7	3.1	4.2	6.5	6.7	16
Wire Gauge	Distance	Distance	Distance	Distance	Distance	Distance
12	590 (180m)	510 (155m)	375 (114m)	245 (75m)	235 (72m)	100 (30m)
10	930 (283m)	810 (247m)	600 (183m)	390 (119m)	575 (175m)	160 (49m)
8	1475 (449m)	1285 (392m)	950 (289m)	615 (187m)	595 (181m)	250 (76m)
6	2350 (716m)	2045 (623m)	1510 (460m)	975 (297m)	945 (288m)	400 (122m)
4	3720 (1134m)	3240 (987m)	2390 (728m)	1545 (471m)	1500 (457m)	630 (192m)

StrongArm Wire Size Chart – 230V Three Phase						
Horsepower	½	¾	1	2	3	5
Amps	2.4	3.0	3.8	6.2	6.4	15.4
Wire Gauge	Distance	Distance	Distance	Distance	Distance	Distance
12	730 (222m)	585 (178m)	460 (140m)	280 (85m)	270 (82m)	115 (35m)
10	1160 (353m)	930 (283m)	730 (222m)	450 (137m)	435 (133m)	180 (55m)
8	1835 (559m)	1470 (448m)	1160 (353m)	710 (216m)	690 (210m)	285 (87m)
6	2925 (891m)	2340 (713m)	1845 (562m)	1130 (344m)	1095 (334m)	455 (139m)
4	4625 (1410m)	3700 (1128m)	2920 (890m)	1790 (546m)	1735 (529m)	720 (219m)

StrongArm Wire Size Chart – 460V Three Phase						
Horsepower	½	¾	1	2	3	5
Amps	1.2	1.5	1.9	3.1	3.2	7.7
Wire Gauge	Distance	Distance	Distance	Distance	Distance	Distance
12	2915 (888m)	2350 (716m)	1850 (564m)	1130 (344m)	1100 (335m)	455 (139m)
10	4640 (1414m)	3710 (1131m)	2930 (893m)	1800 (549m)	1740 (530m)	725 (221m)
8	7340 (2237m)	5870 (1789m)	4650 (1417m)	2840 (866m)	2750 (838m)	1150 (350m)
6	11700 (3566m)	9350 (2850m)	7400 (2255m)	4550 (1387m)	4400 (1341m)	1800 (549m)
4	18500 (5639m)	14800 (4511m)	11700 (3566m)	7200 (2194m)	7000 (2134m)	2900 (884m)